NMC 2012 A-LEVEL PAPER 1 QUESTIONS

SECTION A

Qn1. Without using a calculator, evaluate,

$$S = \sin^2(10^0) + \sin^2(20^0) + \ldots + \sin^2(80^0).$$

Qn2. Given that $|m| - m + n = 10$ and $m + |n| + n = 12$, find the value of $m + n$.

Qn3. Determine the largest possible area of a rectangle with a fixed perimeter of 1000.

Qn4. In a community, two of the three people always lie while the other always tells the truth. A woman asks questions to three men from this community independently . She ask each man; which of you is tallest? John replies, not me. David replies, I am. Geoffrey replies, Not David. Who is tallest?

Qn5. Two particles move clockwise around a circle of circumference 300 cm. The faster particle moves at a constant speed of R cm per second and the slower particle moves at a constant speed of r cm per second. If the particles meet every after 50 seconds, find the difference between R and r.

Qn6. For how many integers x in the set $\{1, 2, 3, \ldots, 99, 100\}$ is $x^3 - x^2$ a square of an integer?

Qn7. Suppose that $f(n) = \log_2 3 \cdot \log_3 4 \cdot \log_4 5 \cdots \log_{n-1} n$, find the value of $\sum_{k=2}^{500} f(2^k)$.

Qn8. A circle passes through two adjacent vertices of a square and is tangent to one side of the square. If the length of the side of the square is $2cm$, find the radius of the circle.

Qn9. A man caught some fish. The two heaviest fish had a combined weight which was 25% of the total weight of all the fish. The five lightest fish had a combined weight which was 45% of the total weight of all fish. He put the two heaviest fish in the freezer and ate the five lightest fish for lunch. His cat took all the remaining fish. How many fish did the cat eat?

Qn10. On Wednesday, four trucks drove in a line (one directly behind the other). None of the trucks bypassed any of the others, so the order of the trucks never changed. How many ways are there to rearrange the order of the trucks so that on the next day, no truck is directly behind a truck that it was directly behind the day before.

SECTION B

Qn11. The equation $x^2 - y^2 = 1,000,001$ has only two solutions where x and y are both in the solution set $S = \{1, 2, 3, 4, \ldots, \}$. Find the smallest value of x in the solution set.

Qn12. A circle is inscribed in a triangle whose sides are 10 cm, 10 cm and 12 cm. A second smaller circle is inscribed tangent to the first circle but also tangent to the two equal sides of the triangle. Find the radius of the second smaller circle.

Qn13. Bill starts walking along a 6km road from the beginning. Sometime later, his brother Phil rides his bike along the same road in the same direction starting at the same point as Bill. Phil can ride three times as fast as Bill can walk. When he passes Bill, Phil on the bike continues to the other end of the road and Bill turns around and heads back towards the starting point. When Phil get to the other end of the road, he turns his bike around and goes back along the same road, arriving at the same time as does Bill. How far from the end of the road did Phil pass Bill?

Qn14. In the Magic square below, the numbers in each row; the numbers in each column and the numbers on each diagonal have the same sum. Given the magic square shown with a, b, c, x, y, z all greater than zero, determine the product xyz in terms of a, b and c.

$\log a$	$\log b$	$\log x$
p	$\log y$	$\log c$
$\log z$	q	r

Qn15. The figure shows a graph of the function of x which models the height of a ramp. The function meets the following requirements:

(i) At $x = 0$, the value of the function is 0 and the slope of the graph of the function is also 0.

(ii) At $x = 4$, the value of the function is 1, and the slope of the graph of the function is 1.

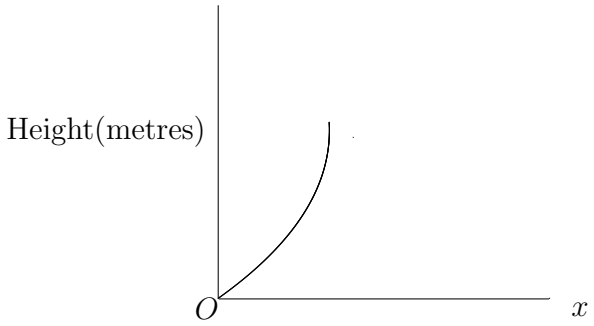
(iii) Between $x = 0$ and $x = 4$ the function is increasing.

(a) Let $f(x) = ax^2$ where a is a nonzero constant. Show that it is not possible to find a value for a so that f meets requirement (ii) above.

(b) Let $g(x) = cx^3 - \frac{x^2}{16}$, where c is a non zero constant. Find the value of c so that g meets requirement (ii) above.

(c) Show that with the value of c obtained in (b) above, the function g does not satisfy requirement (iii) above.

(d) Let $h(x) = \frac{x^n}{k}$ where k is a nonzero constant and n is a positive integer. Find the values of k and n so that h meets requirement (ii) above. Show that h also meets requirements (i) and (iii) above.



NMC 2012 A-LEVEL PAPER 1 SOLUTIONS

SECTION A

Qn1. $S = \cos^2 80^\circ + \cos^2 70^\circ + \cos^2 60^\circ + \cos^2 50^\circ + \sin^2 50^\circ + \sin^2 60^\circ + \sin^2 70^\circ + \sin^2 80^\circ$
$=(\cos^2 80^\circ + \sin^2 80^\circ) + (\cos^2 70^\circ + \sin^2 70^\circ) + (\cos^2 60^\circ + \sin^2 60^\circ) + (\cos^2 50^\circ + \sin^2 50^\circ)$
$= 1 + 1 + 1 + 1 = 4.$

Qn2. $|m| - m + n = 10$ and $m + |n| + n = 12$.

- Case I. When $m > 0, n > 0$, then $n = 10$
and $m + 2n = 12 \Longrightarrow m + 20 = 12 \Longrightarrow m = -8$ which is not possible since m should be positive.

- Case II.When $m > 0, n < 0, |m| - m + n = 10$ and $m = 12$ implying $n = 10$. This is not possible since n should be less than zero.

- Case III. When $m < 0, n < 0$, then $-2m+n = 10$ and $m = 12$ which is not possible again.

- Case IV. When $m < 0, n > 0, -2m + n = 10 \Longrightarrow n = 10 + 2m$ and $m + 2n = 12$ and substituting for n, we get $m + 2(10 + 2m) = 12$
$m + 20 + 4m = 12 \Longrightarrow m = \frac{-8}{5} \Longrightarrow n = 10 + 2 \times \frac{-8}{5} = \frac{34}{5}$ Thus $m + n = \frac{-8}{5} + \frac{36}{5} = \frac{28}{5}$.

Qn3. Let the length be x and width be $y \Longrightarrow A = xy$ and $P = 2(x + y) = 1000$
$\Longrightarrow x + y = 500 \Longrightarrow y = 500 - x$
$A = x(500 - x)$ and $\frac{dA}{dx} = 500 - 2x$.
For max, $500 - 2x = 0 \Longrightarrow 2x = 500$ so that $x = 250$.
$\frac{d^2A}{dx^2} = -2$ since $\frac{d^2A}{dx^2} < 0$. Therefore $x = 250$ and $y = 250$.
The largest possible area is $250 \times 250 = 62500$ square units.

Qn4. John: Not me, David: I am and Geoffrey: Not David
Suppose John tells the truth, then David tells a lie and Geoffrey tells the truth which is a contradiction since two of the three people always tells a lie.
Suppose David tells the truth, it implies John tells the truth too and Geoffrey tells a lie. A contradiction again, so then Geoffrey tells the truth implying that John and David tell a lie. Therefore John is the tallest.

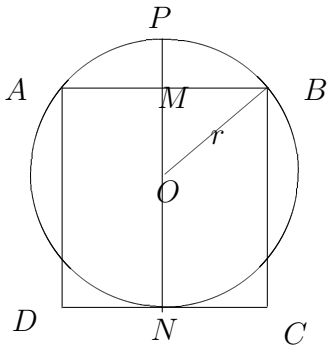
Qn5. Let k be the distance the slower particle travels in 50 seconds. Then $50r = K$ and
$50R = k + 300$. Subtracting the two equations gives $50R - 50r = k + 300 - k$.
Thus $R - r = \frac{300}{6} = 6$ cm per second.

Qn6. Consider $x^3 - x^2 = x^2(x - 1)$, we need $x - 1$ to be a square of an integer and there are exactly ten perfect squares between +1 and 100. Therefore there are 10 integers x in the set $1, 2, 3, ..., 100$ for which $x^3 - x^2$ is a square of an integer.

Qn7. $f(n) = \log_2 3.\log_3 4.\log_4 5...\log_{n-1} n = \frac{\log_2 3}{\log_2 2}.\frac{\log_2 4}{\log_2 3.}.\frac{\log_2 5}{\log_2 4}...\frac{\log_2 n-1}{\log_2 n-2}.\frac{\log_2 n}{\log_2 n-1}.$

$f(n) = \log_2 n.$
$\sum_{k=1}^{500} f(2^k)$ where $f(2^k) = \log_2 2^k.$
$\sum_{k=1}^{500} k = 2 + 3 + 4 + 5 + 6 + ... + 498 + 499 + 500,$
$u_n = a + (n-1)d$ where $u_n = 500, a = 2, d = 1$
$500 = 2 + (n-1)$ which gives $n = 499$. Therefore $S_{499} = \frac{499}{2}(2 \times 2 + (499-1) \times 1) = 125249.$

Qn8. Let $ABCD$ be the vertices of a square. Let M be the midpoint of the segment AB. let r be the radius of the circle. Given $\overline{AB} = 2cm$, then $\overline{MB} = 1cm.\overline{MN} = 2cm$ and $\overline{PN} = 2rcm$. But $\overline{MP} = \overline{PN} - \overline{MN} = (2r-2)cm$. But $\overline{OP} = r$ so $\overline{OM} = \overline{OP} - \overline{MP} = r - (2r-2) = (2-r)cm$.



Using triangle OMB, $\overline{OM}^2 + \overline{MB}^2 = \overline{OB}^2 \Longrightarrow (2-r)^2 + 1^2 = r^2$
$4 - 4r + r^2 + 1 = r^2 \Longrightarrow 5 - 4r = 0$ so that $r = \frac{5}{4}$. The radius of the circle is $\frac{5}{4}cm$.

Qn9. Let W be the total weight of all fish the man caught. The fish taken by the cat had weight of 30% .

- If the cat took 4 fish or more, their average weight must have been at most $7.5\%W(\frac{30}{4} = 7.5)$ which is less than the average weight of the 5 lightest fish.
- If the cat took 1 or 2 fish, then their average weight must have been atleast $15\%W(\frac{30}{2})$ which is more than the average weight of the 2 heaviest fish.

Therefore the cat took 3 fish.

Qn10. Suppose we label the trucks $1, 2, 3, 4$ with the labels corresponding to the position of the trucks on Wednesday. There are 11 possible arrangements the next day. $1324, 1432, 2143, 2413, 2431, 3142, 3214$
Hence there are 11 ways.

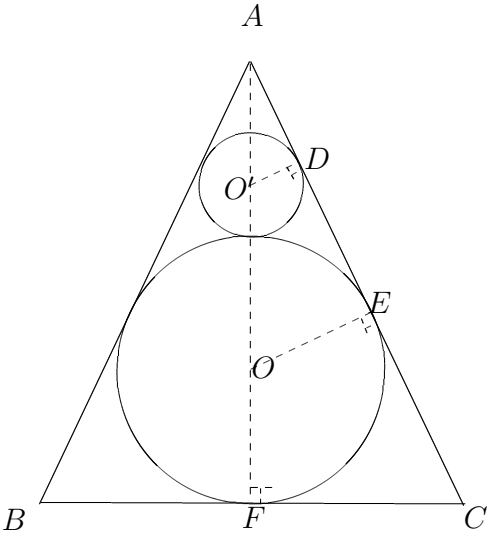
SECTION B

Qn11. $x^2 - y^2 = 1000001$ can be written as $x^2 - y^2 = 1000000 + 1 = 10^6 + 1$
$(x-y)(x+y) = 1(10^6+1)$ so that $x - y = 1 \Longrightarrow x = y + 1.$
$x + y = 1000001(10^6+1) \Longrightarrow y + 1 + y = 1000001 \Longrightarrow 2y = 1000000 \Longrightarrow y = 500000.$
Also $x^2 - y^2 = 10^6 + 1$, But $10^6 + 1 = (10^2+1)(10^4 - 10^2 + 1)$
$(x-y)(x+y) = (101)(9901).$
Thus $x-y = 101$ and $x+y = 9901$ and on solving simultaneously gives $x = 5001$ and $y = 4901.$
The smallest value of x is 5001.

Qn12. $\overline{CF} = \overline{CE} = 6cm$ since they are tangents to the circle. Thus $\overline{OF} = \overline{OE}$. Since $\overline{AC} = 10cm$ and $\overline{CE} = 6cm \Longrightarrow \overline{AE} = 4cm$.
Using right angled triangle AFC, $AF^2 + FC^2 = AC^2 \Longrightarrow AF^2 = 10^2 - 6^2 = 64$ so that $AF = 8$cm.
Triangles AEO and AFC are similar, so $\frac{AF}{AE} = \frac{FC}{OE}$ and on subtraction $\frac{8}{4} = \frac{6}{OE} \Longrightarrow OE = 3cm$.

Hence $\overline{OE} = \overline{GO} = \overline{OF} = 3cm$ which is the radius of the circle.
$\overline{AG} = \overline{AF} - \overline{GF} = 8 - 6 = 2cm$. Also $O'A = AG - O'G = 2 - r$ since $\overline{O'D}$ is parallel $\overline{OE}$, then triangles ADO' and AEO are similar and $\frac{O'D}{O'A} = \frac{OE}{OA}$.

But AEO is a right angled triangle and $\overline{AE} = 4cm, \overline{OE} = 3cm, \overline{AO} = 5cm$.
Thus by substitution $\frac{r}{2-r} = \frac{3}{5}$ so that $r = \frac{3}{4} = 0.75cm$.



Qn13. Let the speed of Bill be x. Then the speed of Phil is $3x \Longrightarrow$ the ratio of their speeds is $3:1$ respectively. The total time that Bill uses for the whole journey is $\frac{12}{3x} = \frac{4}{x}$.
Let t_1 be the time that Phil waited before he set off after Bill had left, then
$-t_1 + \frac{2(6-d)}{x} = \frac{4}{x}$ where $\frac{2(6-d)}{x}$ is the time taken for Bill to cover to and from the meeting point.
From the previous equation $-t_1 = \frac{4}{x} - \frac{12}{x} + \frac{2d}{x} \Longrightarrow -t_1 = \frac{-8}{x} + \frac{2d}{x}$
$t_1 = \frac{8-2d}{x}$.
Thus, Bill will have moved a distance of $8 - 2d$ when Bill starts and the remaining distance of $6 - (8 - 2d) = 2d - 2$.

Bill will cover $\frac{2d-2}{4}$ up to the meeting point and Phill will cover $\frac{3(2d-2)}{4}$, so the total distance

covered by Bill when Phil is also moving is $\frac{2(2d-2)}{4} + 8 - 2d = d - 1 + 8 - 2d = 7 - d$.
Using ratios, Distances $(7-d) : 12$ and the speeds $1:3$.
Thus $\frac{7-d}{12} = \frac{1}{3} \Longrightarrow 3(7-d) = 12 \Longrightarrow 9 = 3d$
$d = 3km$.

Qn14. let R_1, R_2, R_3 represent the three rows, and C_1, C_2, C_3 three columns, D_1 the diagonal from the top left to the top right and D_2 the diagonal from the top left to bottom right.

Sum of R_1= sum of D_1.
$\log a + \log b + \log x = \log z + \log y + \log x$.
This implies $\log ab = \log yz \Longrightarrow ab = yz$ or $z = \frac{ab}{y}$.
Also sum of C_1 = sum R_2.
$\log a + \log p + \log z = p + \log y + \log c$
$az = cy$ or $z = \frac{cy}{a}$.

since $z = \frac{ab}{y}$ and $z = \frac{cy}{a} \Longrightarrow \frac{ab}{y} = \frac{cy}{a}$ or $y^2 = \frac{a^2b}{c}$

$y = \sqrt{\frac{a^2b}{c}}$ or $y = \frac{a\sqrt{b}}{\sqrt{c}}$.

sum of C_3 = sum of D_2
$\log x + \log c + r = \log a + \log y + r$
$\log xc = \log ay \Longrightarrow x = \frac{ay}{c}$.

Thus $xyz = \frac{ay}{c}\frac{y}{1}\frac{cy}{a} = y^3$. Therefore $xyz = \frac{a^3b^{\frac{3}{2}}}{c^{\frac{3}{2}}}$.

Qn15. (a) $f(x) = ax^2 \Longrightarrow f(4) = 1 = a.4^2 \Longrightarrow a = \frac{1}{16}$. Since the slope of the function is 1,

$\Longrightarrow \frac{df}{dx} = 2a(4) = 1$.
$a = \frac{1}{8}$. Thus f cannot satisfy (ii) since $\frac{1}{16} \neq \frac{1}{8}$.

(b) $g(x) = cx^3 - \frac{x^2}{16} \Longrightarrow g(4) = 64c - 1 = 1$
$c = \frac{1}{32}$.

When $c = \frac{1}{32}$, $\frac{dg}{dx} = 3cx^2 - \frac{2x}{16}$.
Therefore $g'(4) = 3 \times \frac{1}{32} \times 16 - 2(\frac{1}{2}) = 1$.

(c) $g'(x) = \frac{3x^2}{32} - \frac{x}{8} = \frac{x(3x-4)}{32}$.

- when $x < 0$, $g'(x) > 0$, so $g(x)$ is increasing.
- When $0 < x < \frac{4}{3}$, $g'(x) < 0$. and so $g(x)$ is decreasing.
- When $\frac{4}{3} < 0$, $g'(x) > 0$ and so $g(x)$ is increasing.

This means $g(x)$ does not increase only when $0 < x < \frac{4}{3}$.
$g'(x) < 0$ for $0 < x < \frac{4}{3}$ so g does not satisfy condition (iii).

(d) $h(x) = \frac{x^n}{k} \Longrightarrow h(4) = \frac{4^n}{k} = 1 \Longrightarrow 4^n = k$.

$h'(x) = \frac{nx^{n-1}}{k}. \Longrightarrow h'(4) = \frac{n4^{n-1}}{k} = \frac{n4^{n-1}}{4^n} = \frac{n}{4} = 1$. Hence $n = 4$ and $k = 4^4 = 256$.

$h(x) = \frac{x^4}{256} \Longrightarrow h(0) = 0.$

$h'(x) = \frac{4x^3}{256} \Longrightarrow h'(0) = 0$ and $h'(x) > 0$ for $0 < x < 4$.
Therefore $h(0) = 0, h'(0) = 0$ which meets requirement (i).
$h'(x) = \frac{4x^3}{256} = \frac{x^3}{64}$. This implies that $x > 0$ and $h'(x) > 0. \Longrightarrow h(x)$ is increasing and this meets the requirement (iii).